CONFIRMING THE NULL: REMARKS ON EQUIVALENCE TESTING AND THE TOPOLOGY OF CONFIRMATION

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ABSTRACT. Null Hypothesis Statistical Testing is a dominant framework for conducting statistical analysis across the sciences. There remains considerable debate as to whether, and under what circumstances, evidence can be said to be confirmatory of a null hypothesis. This paper presents a modal logic of short-run frequentist confirmation developed by leveraging the duality between hypothesis testing and statistical estimation.

It is shown that a hypothesis is confirmable if and only if it satisfies the topological condition of having nonempty interior. Consequently, two-sided hypotheses are not statistically confirmable owing to defects in their topological structure. Equivalence hypotheses are, by contrast, confirmable.

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1. Is there a Frequentist Theory of Short-Run Confirmation?

The question of whether evidence can be in support of or confirmatory of a scientific theory is of critical importance in the philosophy and interpretation of science. Scientific theories often guide actual decisionmaking, so having a theory of evidentiary support in favor of a theory is critical. A highly developed candidate for confirmation theory is given by Bayesian confirmation theory [1]. Confirmation within the paradigm of frequentist statistics is, by contrast, far from settled. There is a lack of consensus as to whether frequentist statistics admits a formal theory of confirmation and, if it does, what the necessary and sufficient conditions for confirmation of a hypothesis are.

By some accounts, hypotheses are never confirmed, only retained or rejected according to a default logic of hypothesis testing [6]. On other accounts, such as those of Frick [2] and Mayo [8], a hypothesis is only to be accepted or considered severely tested just in case a good-faith effort to refute the hypothesis with a high chance of detecting an error in the hypothesis, if indeed there was one, was performed. By contrast, Lakens [5] has recently claimed that two-sided hypotheses are not confirmable.

In the present paper I develop a modal logic of evidentiary support or confirmation in the short run, i.e. for finite sets of evidence E. In this logic, there are three possible results of a test of a hypothesis H: H can be confirmed, rejected, or the test can be simply inconclusive. This theory depends crucially on the duality between hypothesis testing and estimation using confidence regions. On this account, the result of a statistical test is determined by the set-theoretic relationship between a hypothesis H and a $(1-\alpha)$ confidence region R_α: a hypothesis is confirmed just in case $R_\alpha \subseteq H$, rejected if $R_\alpha \subseteq H^c$, and is inconclusive if R_α nontrivially intersects both H and its complement H^c. The criteria for whether a hypothesis is in-principle confirmable can be characterized by a simple topological property of the hypothesis H, namely that H contains nonempty interior. In particular, two-sided hypotheses of the form $H : \theta = \theta_0$ in topological spaces such as $\mathbb{R}$ are not confirmable, while equivalence and non-inferiority hypotheses of the form $H : \theta \in [L, U]$ are confirmable. This suggests that the standard two-sided hypotheses are unduly emphasized within scientific practice: while refutable, they are not confirmable.

Prior work on the relationship between the topological structure of a hypothesis H and its statistical verifiability was done by Genin and Kelly, who gave a topological characterization of hypotheses that are guaranteed to be confirmed if true [3]. By contrast, this work focuses on characterizing hypotheses which, in principle, are statistically confirmable by *some* data.

Finally, the logic of confirmation developed in this paper is used to give a formal semantics for Mayo's notion of severe testing and the Full Severity Principle. Mayo has argued in favor of a "meta-statistical principle" that "ensure that only statistical hypotheses that have passed severe or probative tests are inferred from the data" by way of criterion of severity that "avoid[s] classic fallacies from tests that are overly sensitive, as well as those not sensitive enough to particular errors and discrepancies" [10]. The fallacies Mayo speaks of are an artifact of attempting to confirm two-sided hypotheses. On my account, a theory τ is (partially) confirmed just in case an associated hypothesis H_τ expressing the empirical adequacy of the theory is confirmed.

2. Doxastic Interpretations of Hypothesis Testing

Null Hypothesis Statistical Testing (NHST) is a dominant mode of scientific inference. In NHST, given a prespecified significance level α the standard way to interpret an observed p-value $> \alpha$ for a null hypothesis H_0 is given the gloss of "retaining the null H_0" or "failure to reject H_0" at level α. We assume that Θ is a parametric family of probability distributions on sample space Ω, $H_0 \subset \Theta$ is a set of parameters, and that the alternative hypothesis $H_1 = H_0^c$ is the complement of H_0 within Θ. We further assume that evidence E is drawn independently and identically distributed from some distribution $\theta \in \Theta$. For each n, this data uniquely determines a probability measure $\mathbb{P}_\theta$ on Ω^n given by $\theta^{\otimes n}$.

2.1. **The Default Account of Hypothesis Testing.** Lehmann/Romano [6] is a standard, orthodox frequentist reference for hypothesis testing. For them, hypothesis testing is a tool to solve the following decision problem: on the basis of collected evidence E, should we *accept* or *reject* the hypothesis? As they describe it:

> We now begin the study of the statistical problem that forms the principal subject of this book, the problem of hypothesis testing. As the term suggests, one wishes to decide whether or not some hypothesis that has been formulated is correct. The choice here lies between only two decisions: accepting or rejecting the hypothesis. A decision procedure for such a problem is called a test of the hypothesis in question. (Page 61)

Salient to this framing of hypothesis testing are the following features:

(1) (Asymmetric Roles of Null and Alternative Hypotheses) There is a specified hypothesis H_0, the *null* hypothesis, that will be accepted *unless overwhelming evidence overrides it,* and
(2) (Bivalent Decision Problem) There are only two possible values for the decision problem: *accept* or *reject*.

The null hypothesis H_0 serves as the *default* output of the test: if the test does not give sufficient evidence against H_0, H_0 is accepted. For Lehmann and Romano, the design of the test of H_0 should be related in some way to the investigator's attitude regarding the truth or plausibility of H_0, and should in part be calibrated as a function of the level of evidence required to shake their belief in the hypothesis:

> Another consideration that may enter into the specification of a significance level is the attitude toward the hypothesis before the experiment is performed. If one firmly believes the hypothesis to be true, extremely convincing evidence will be required before one is willing to give up this belief, and the significance level will accordingly be set very low. (Page 63)